

MATLAB projects

I am interested in MATLAB-based projects. Please list the projects that EFY has published on this topic.

Arshad Shaik

Belgaum

EFY: Listed below are software projects recently published in EFY:

1. 555 timer design using MATLAB software, published in July 2013 issue of EFY

2. Graphics object counter simulation using MATLAB software (January 2013)

3. Huffman encoding and decoding in MATLAB (November 2012)

4. Gain and noise calculations of cascaded systems using MATLAB (September 2012)

5. Artificial neural network simulation in Labview and MATLAB (August 2012)

6. MATLAB implementation of antenna design and applications software (May 2012)

7. Digital filter design using MATLAB (February 2012)

8. Multifunction unit converter and calculator using MATLAB (December 2011)

9. Visitor counter using MATLAB circuit (November 2011)

10. Parallel port controller using MATLAB circuit (May 2009)

11. Robotic vision using MATLAB (May 2009)

EFY DVD

I got EFY August issue along with the DVD. I want to know how to use Altium Designer and LaTeX, both provided on the DVD.

Madhu Prathik

Coimbatore

EFY: After downloading the Altium Designer into your system, you will find the folder, 'Tutorial-Getting Started with PCB Design.pdf,' with step-by-step details on how to use the software. The LaTeX markup language is included under the tutorial section in the DVD. You can find more details on how to use it after installing the editor. The LaTeX editor can be downloaded from <http://www.latexeditor.org/>

'Spot An Error' Award Winners

In the parts list of the construction project, 'Environment Monitoring System Using Arduino' published in the August 2013 issue of EFY, the value of R2 is mistakenly printed as 100 ohm instead of 20 kilo-ohm.

Somnath Biswas

□ In the 'Transistor Polarity Checker' circuit idea published in August 2013, diodes D3 and D4 are wrongly biased; one of them should be connected in reverse biased mode.

Jayati

□ In the 'Buyers Guide: Signal Generator' published in the August 2013 issue, in column three of Page 90. In the sentence that refers to the figure: "It is inversely related to the wavelength (shown in Fig 3 as point 4)"...it is not correct. It should be Fig 2 instead of Fig 3.

Sammuihdhin

□ In the same article, in Fig.4, the rise-time is written twice; the second one should be fall-time.

Yuvaraja Ravi

Microcontroller-based projects

I want microcontroller-based projects. Kindly let me know from where I can get these projects.

Avani Patel

Through email

EFY: We have published many microcontroller-based projects in EFY. You can get the book, 'Microcontroller Based Projects' (first edition) from our associates Kits'n'Spares (info@kitsnspares.com). Its second edition is available online in PDF format.

Also, the complete kits for some of these projects are available with Kits'n'Spares.

Knowing more about footprints in gEDA

I read a question in the May issue of EFY about gEDA, where you referred to an article in the January issue of EFY. I have read the article and want to know more about footprints. I know what they are but don't know which footprints to use in gschem, i.e., what the codes like ACY400 mean and what codes exist. Is there any handbook or any other reference to use? This has been the main thing stopping me from using the PCB design tool. Eagle has a preview that lets you know what footprint to use. That's what I've been using so far. Thank you for your time!

Ashwith Rego

Bengaluru

EFY: We started implementing gEDA project when we knew almost nothing about it and then completely customised it to our requirements as a publishing house. During the course of this process, we understood that the biggest strength of this tool is its customisability. The same applies to the PCB design tool too.

Here are some simple steps that can make things more comfortable for you:

1. To successfully import your schematic created in gschem into the PCB tool, you need to correctly enter the footprint name in the attributes for each component.

2. If you don't see a footprint attribute as shown by the red box in Fig. 1, select it from the drop-down menu shown by the green box and enter the footprint name in the value field.

3. Press the 'Add' button. The same needs to be repeated for each component.

4. Now, the question is how to know which footprint name needs to be entered without a preview plane. For that, just keep the PCB tool open in parallel to gschem. Open 'Library' using the 'Window' tab at the top.

5. The parts in the library can be viewed in the preview plane as shown in Fig. 2.

6. Now, the name of your interest that you need to put in the attributes is normally in square brackets, e.g.,